

TIPE

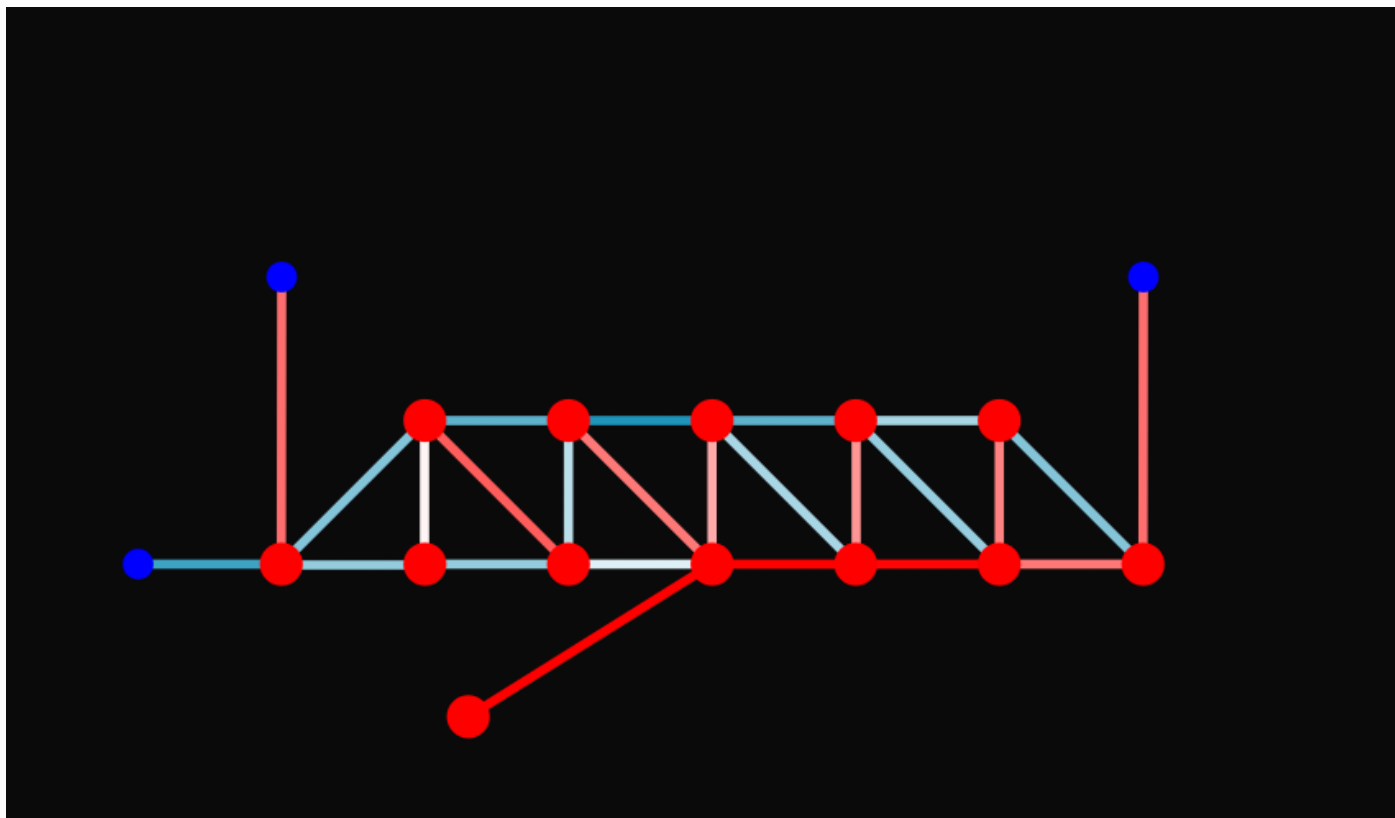
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June 22, 2025 — Lycée Pierre Corneille, Rouen



Simulation Physique



Contraintes sur un pont



Calculs

$$\dot{C} : \frac{dC}{dt} = \frac{\partial C}{\partial q} \dot{q}$$

$$\ddot{C} : \dot{J}\dot{q} + J\ddot{q} = 0$$

$$\mathcal{P}(F_c) = F_c \cdot \dot{q}$$

On remarque alors que :

$$\forall \lambda \in \mathbb{R}, (J^T \lambda) \cdot \dot{q} = (J^T \lambda)^T \dot{q} = \lambda J \dot{q} = 0$$

Accélération :

$$\ddot{C} = 0 \Leftrightarrow J\ddot{q} + \dot{J}\dot{q} = 0$$

$$\Leftrightarrow J\ddot{q} = -\dot{J}\dot{q}$$

$$\Leftrightarrow JW(F_c + F) = -\dot{J}\dot{q}$$

$$\Leftrightarrow JWF_c = -JWF - \dot{J}\dot{q}$$

$$\Leftrightarrow JWJ^T \lambda = -JWF - \dot{J}\dot{q}$$

Jacobienne : $J = \frac{q^T}{\|q\|}$

Vitesse :

$$\dot{C} = 0 \Leftrightarrow J\dot{q} = 0$$

$$\Leftrightarrow JW \Delta t J^T \lambda = -J\dot{q}$$

$$\Leftrightarrow JWJ^T \lambda' = -J\dot{q}$$

Contrainte sur la position :

$$C : \|q\| - l = 0$$

Contrainte sur la vitesse :

$$\frac{dC}{dT} = 0 \Leftrightarrow \frac{q^T}{\|q\|} \dot{q} = 0$$

Contrainte sur l'accélération :

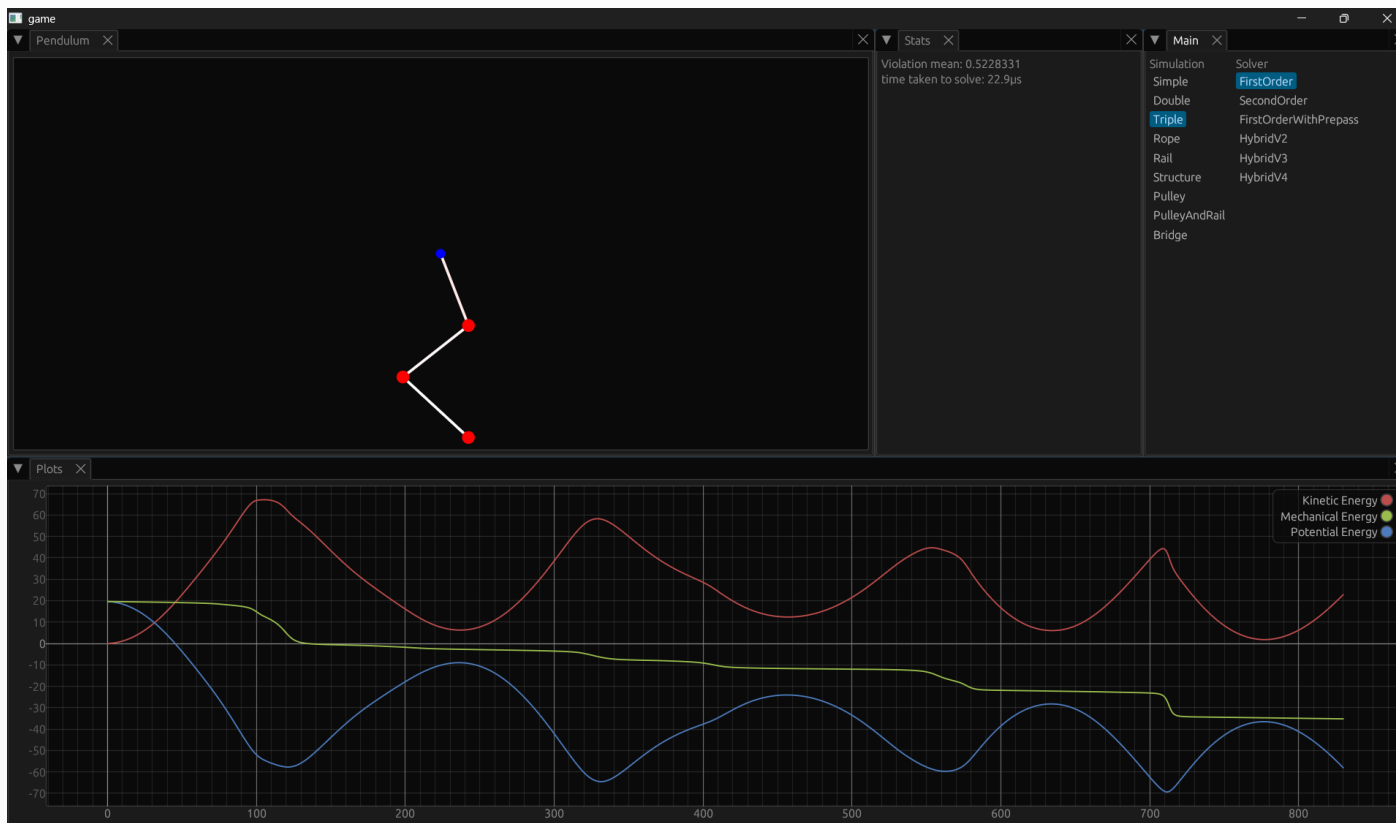
$$\frac{d^2 C}{dt^2} = 0 \Leftrightarrow \frac{\dot{q}^2 q^2 - (q \cdot \dot{q})^2}{\|q\|^3} + \frac{q}{\|q\|} \cdot \ddot{q} = 0$$

Second terme ordre à l'ordre 2 :

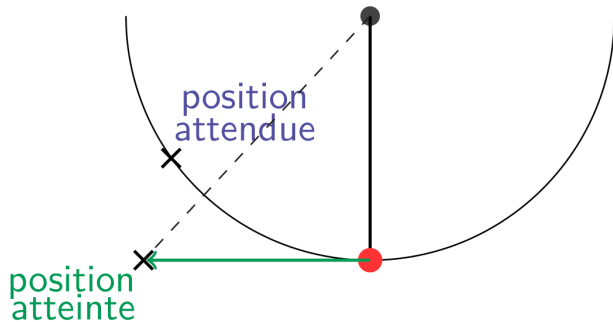
$$\dot{J}\dot{q} = \frac{\dot{q}^2 q^2 - (q \cdot \dot{q})^2}{\|q\|^3} = \frac{(x\dot{y} - y\dot{x})^2}{(x+y)^{\frac{3}{2}}}$$



Pertes d'énergie



Correction de trajectoire



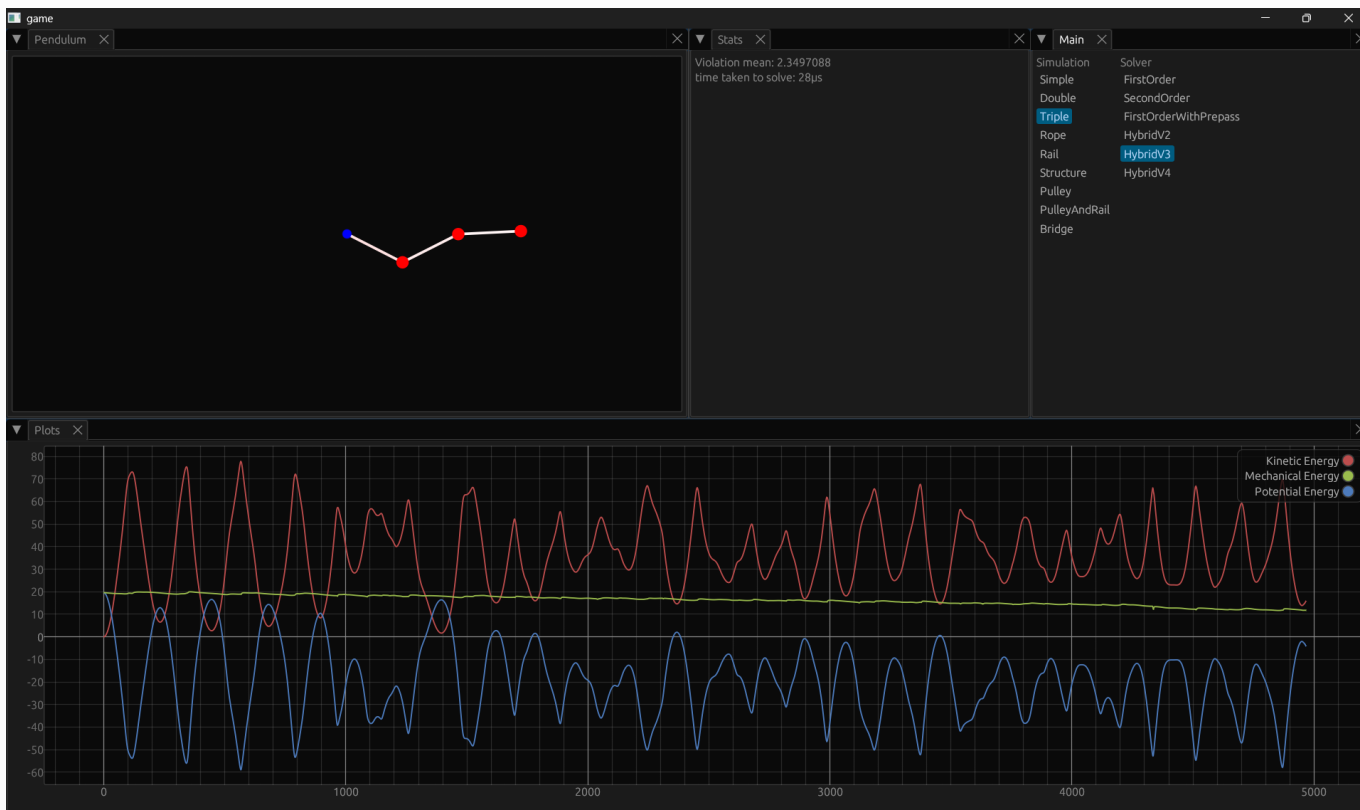
$$q(t+\Delta t) = q(t) + \dot{q}(t)\Delta t + \frac{\ddot{q}(t)}{2}\Delta t^2 + O(\Delta t^3)$$

$$\frac{q(t+\Delta t) - q(t)}{\Delta t} = \dot{q}(t) + \frac{\ddot{q}(t)}{2}\Delta t + O(\Delta t^2)$$

$$JWJ^T\lambda' = -J\dot{q} - \frac{1}{2\Delta t}J\ddot{q}$$



Correction du solveur





Simulation Fluide



Equations de Navier-Stokes

Équation de Navier-Stokes (fluide incompressible)

$$\begin{cases} \frac{\partial \mathbf{u}}{\partial t} = -\mathbf{u} \cdot \nabla \mathbf{u} + \mathbf{g} + \nu \Delta \mathbf{u} - \frac{1}{\rho} \nabla p \\ \nabla \cdot \mathbf{u} = 0 \end{cases}$$



Equations de Navier-Stokes

Équation de Navier-Stokes (fluide incompressible)

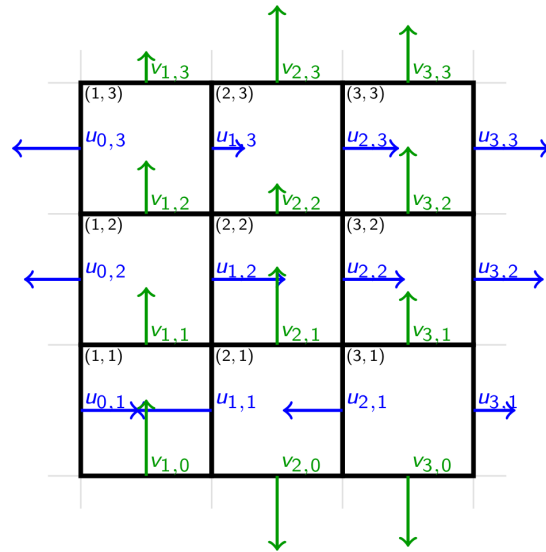
$$\begin{cases} \frac{\partial \mathbf{u}}{\partial t} = -\mathbf{u} \cdot \nabla \mathbf{u} + \mathbf{g} + \nu \Delta \mathbf{u} - \frac{1}{\rho} \nabla p \\ \nabla \cdot \mathbf{u} = 0 \end{cases}$$

Accélération ($\frac{\partial \mathbf{u}}{\partial t}$) = **Advection** ($-\mathbf{u} \cdot \nabla \mathbf{u}$) + **Forces externes** ($\mathbf{g}$) + **Viscosité** ($\nu \Delta \mathbf{u}$) - **Pression** ($\frac{1}{\rho} \nabla p$)

Incompressibilité ($\nabla \cdot \mathbf{u} = 0$) : pas d'ajout ou de retrait de fluide

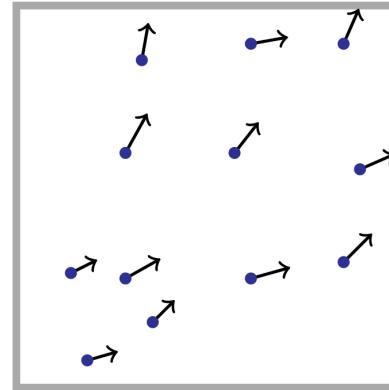
Choix de la méthode de représentation

Eulérien (Grille décalée)



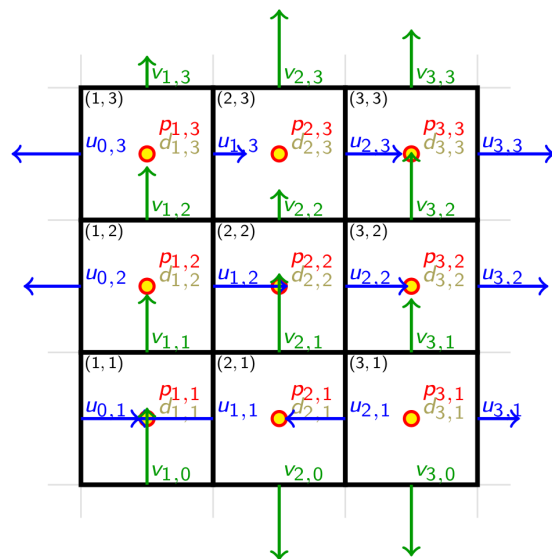
- Vitesse $u_{i,j}$ (face verticale)
- ↑ Vitesse $v_{i,j}$ (face horizontale)
- (i,j) Indices de la cellule

Lagrangien (Particules)



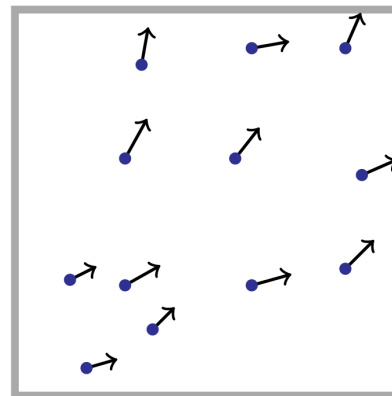
Choix de la méthode de représentation

Eulérien (Grille décalée)

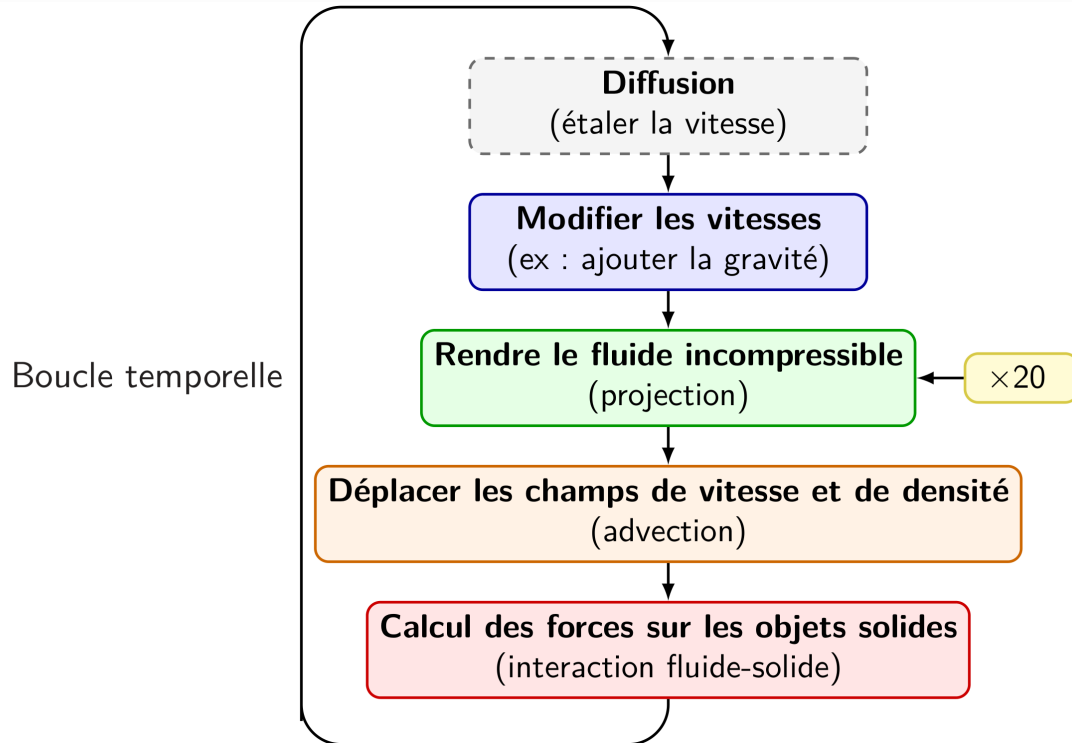


- Pression $p_{i,j}$
- Densité $d_{i,j}$
- Vitesse $u_{i,j}$ (face verticale)
- ↑ Vitesse $v_{i,j}$ (face horizontale)
- (i,j) Indices de la cellule

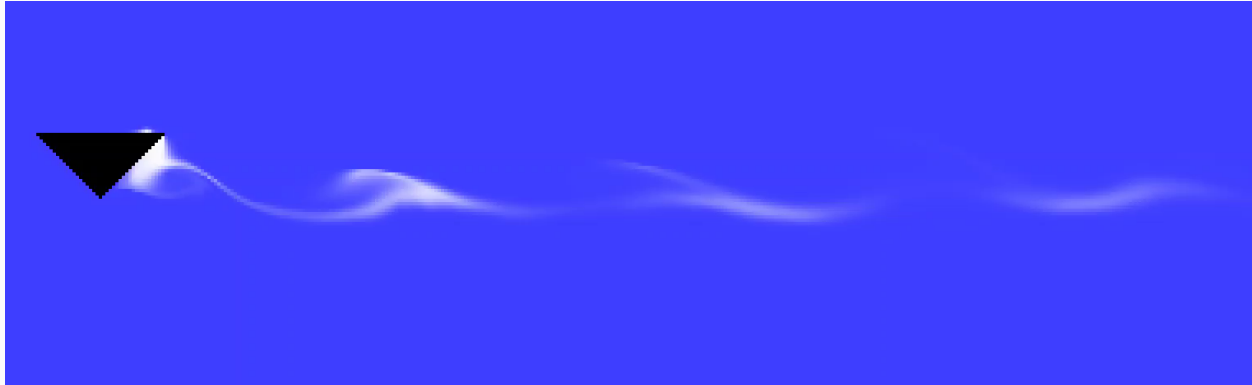
Lagrangien (Particules)



Fonctionnement du code



Vortex de Von Karman

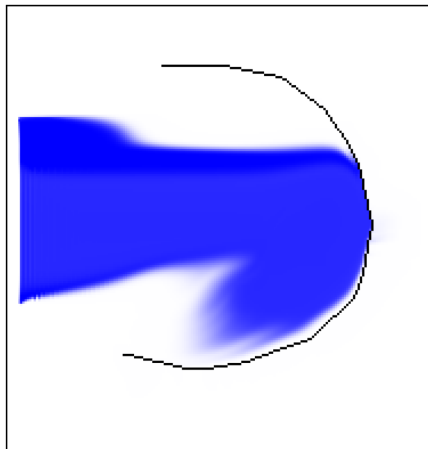
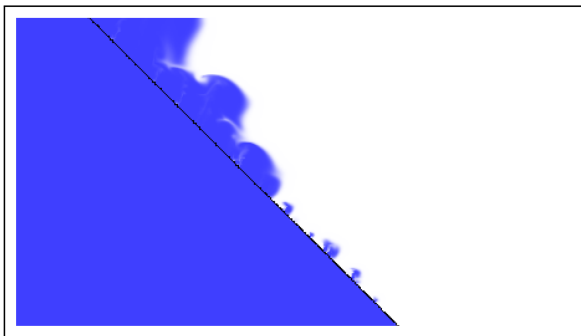


Nombre de Reynolds (Re):

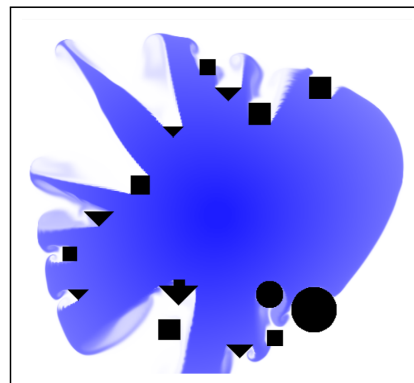
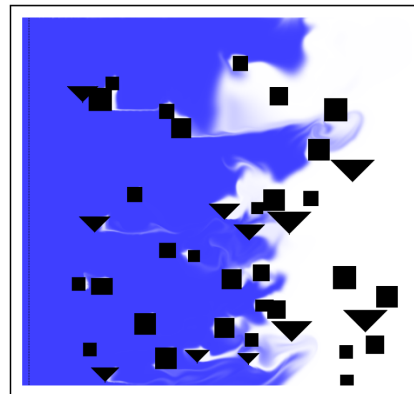
$$Re = \frac{\rho \| [\vec{v} \cdot \overrightarrow{grad}] \vec{v} \|}{\eta \| \Delta \vec{v} \|}$$

Pour notre part, on obtient un nombre de Reynolds de 50,7.

Problèmes



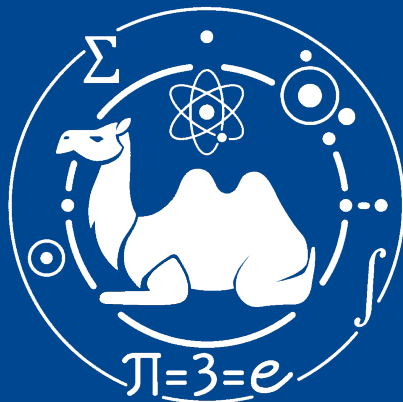
Exemples





Objectifs :

- Régler les problèmes de conservation de la masse du fluide
- Améliorer la précision du solveur fluide
- Passer a un univers en 3D et infini
- Etudier les méthodes de protection des structures face aux vents



—•MP2I•—

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2024 - 2025